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Advanced Static Model Screening Using Non-Supervised Machine Learning Techniques

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Effective selection of static reservoir models is a critical step in enhancing the accuracy, efficiency, and reliability of reservoir simulation workflows. Traditional approaches often rely on volumetric properties alone, which may not fully capture the dynamic behaviour of the reservoir under production. This study introduces a novel methodology that integrates streamline simulation outputs into the model selection process, offering a more performance-oriented perspective. By applying Principal Component Analysis (PCA) to streamline-derived metrics—such as sweep efficiency, flow connectivity, and drainage patterns—the method reduces the dimensionality of the dataset while preserving the most informative features. Hierarchical clustering is then used to group realizations based on their dynamic similarity, enabling a structured and data-driven classification of the model ensemble.

A key innovation of this approach lies in its multi-scale integration of field-level and well-level data, which allows for a more comprehensive understanding of reservoir response. Field-level metrics provide insight into global reservoir performance, while well-level indicators capture localized dynamics such as individual well contributions, connate water production, and breakthrough timing. By combining these two scales of analysis, the methodology ensures that selected models reflect both the broader reservoir trends and the nuanced output pattern of specific wells. This dual-layered perspective enhances the fidelity of reservoir characterization and supports more informed decision-making, particularly in complex or heterogeneous reservoirs where local variations can significantly impact overall performance.

To support interpretation and streamline the selection process, an interactive visualization dashboard is employed. This tool enables real-time exploration of the clustered ensemble, with each realization color-coded by its assigned cluster label. Users can dynamically filter, compare, and validate models, facilitating the generation of a shortlist of representative cases for full-physics simulation. By focusing on a reduced set of high-impact realizations, the workflow significantly lowers computational costs and accelerates simulation timelines—without compromising the diversity or representativeness of the model set. Ultimately, this integrated methodology offers a powerful and scalable solution for optimizing reservoir development strategies, improving operational planning, and enabling more agile, data-driven reservoir management.

Introduction

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Limitations of Uncertainty Analysis in Static Grid Models

Uncertainty analysis is a cornerstone of reservoir modelling, aimed at quantifying subsurface variability and supporting risk-informed decision-making. However, when such analysis is restricted to static grid models—without evaluating the dynamic response of the reservoir—the scope and reliability of the insights derived are significantly limited.

Static models primarily encapsulate geological and petrophysical heterogeneities, such as porosity, permeability, and structural configuration. While these parameters are essential for estimating volumes in place and defining reservoir architecture, they do not capture the dynamic interactions that govern fluid flow and production performance. Consequently, uncertainty assessments based solely on static realizations may fail to account for critical factors such as reservoir connectivity, pressure support, and sweep efficiency, which are only observable through dynamic simulation.

A key challenge lies in the inability to translate static heterogeneities into performance metrics. For example, two static models with similar volumetric properties may exhibit vastly different production profiles due to variations in flow paths, well placement sensitivity, or pressure communication. Without simulating these dynamic responses, it is impossible to quantify the impact of geological uncertainty on production forecasts, recovery factors, or operational risks (Caers, 2011). Moreover, static-only uncertainty analysis limits the ability to identify representative scenarios for decision-making. In the absence of dynamic simulation, clustering or ranking of realizations based on performance metrics cannot be performed, thereby

reducing the analytical value of the model ensemble. This limitation is particularly critical in early field development stages, where fewer well penetrations and sparse data exacerbate uncertainty.

According to Bentley (2020), the integration of static and dynamic modelling is essential for quantifying uncertainty in subsurface interpretations and predictions. Static models alone may misrepresent flow response due to oversimplified assumptions or lack of connectivity analysis. Dynamic modelling helps overcome these limitations by simulating realistic reservoir responses, thus validating the geological scenarios and refining the uncertainty envelope. Ringrose and Bentley (2021) also advocate for model-based uncertainty handling that spans both static and dynamic domains. Their work highlights the importance of ensemble modelling, where multiple static realizations are passed through dynamic simulators to generate a distribution of outcomes. This approach supports scenario ranking, representative case selection, and risk-based decision-making. Hence, static models alone cannot capture the temporal evolution of reservoir performance, nor can they assess the impact of geological uncertainty on production forecasts. Therefore, coupling static uncertainty analysis with dynamic modelling is essential for a comprehensive understanding of reservoir dynamics.

In summary, while static grid models provide an essential understanding of subsurface variability, proper uncertainty analysis requires coupling these models and assumptions with dynamic simulations. This integration enables a more comprehensive assessment of reservoir behaviour, supports robust scenario selection, and enhances the reliability of forecasting and optimization workflows.

Practical Considerations in Dynamic Validation

While dynamic simulation is strongly recommended for validating static grid uncertainty analysis and capturing the full spectrum of reservoir flow dynamics, it is important to recognize the practical constraints associated with this approach. Running dynamic simulations across a large ensemble of static realizations can be computationally intensive and time-consuming, particularly for complex reservoir models with fine-scale resolution and long production timelines.

In many cases, the cost of performing full-physics simulations for every static scenario may be prohibitive, especially during early project phases or when rapid decision-making is required. As a result, practitioners often resort to scenario reduction techniques—such as clustering and representative case selection—to limit the number of dynamic runs while preserving geological diversity and performance variability.

This trade-off between accuracy and efficiency underscores the need for smart integration strategies, where static uncertainty is explored broadly, and dynamic validation is applied selectively to high-impact or representative cases. Such approaches help balance computational resources with analytical rigor, enabling informed decisions without incurring excessive simulation overhead.

Streamline Simulation for Efficient Screening of Geological Realizations

Streamline simulation offers a computationally efficient alternative to full-physics dynamic modelling, enabling the rapid evaluation of fluid flow variability across multiple geological realizations. While it simplifies certain physical processes, streamline simulation retains key dynamic features such as flow paths and well-reservoir interactions, making it particularly valuable for early-stage screening and uncertainty analysis.

By applying streamline simulation to an ensemble of static models, it becomes possible to assess similarities and differences in dynamic response—such as sweep efficiency, inter-well connectivity, and drainage patterns—without incurring the high computational cost of traditional simulators. This approach allows for the identification of clusters of realizations with comparable flow response, supporting representative case selection and scenario reduction.

In essence, streamline simulation can bridge the gap between static uncertainty and dynamic validation, offering a practical tool for incorporating flow-based insights into geological screening workflows. It enhances the interpretability of static models and provides a foundation for more targeted and efficient dynamic simulation efforts.

Objective

The primary objective of this study is to evaluate the impact of connectivity variability on the estimation of recoverable hydrocarbon volumes within reservoir models. Connectivity—defined by the spatial and dynamic relationships between high-permeability zones, flow paths, and well locations—is a critical factor influencing fluid movement and production efficiency. Variations in connectivity across geological realizations can lead to significant differences in reservoir performance, even when volumetric properties appear similar.

To investigate this, streamline simulation techniques are employed across a diverse ensemble of static geological models. As mentioned in previous sections, streamline simulation offers a computationally efficient means of capturing essential dynamic behaviours such as flow directionality and inter-well communication, without the full computational burden of traditional finite-difference simulators. This enables the rapid screening of flow responses and connectivity patterns across a considerable number of realizations.

The dynamic outputs from streamline simulations—such as sweep efficiency, drainage volumes, and well-pair connectivity—are used as input features for a hierarchical clustering process. This unsupervised learning technique groups realizations based on similarity in their dynamic behaviour, revealing structural patterns and performance trends within the ensemble. The clustering results facilitate the identification of representative models that encapsulate the dominant geological and flow characteristics of each group.

By selecting a reduced set of representative cases from the clustered ensemble, the study enables a lean simulation strategy for subsequent full-physics reservoir modelling. This targeted approach significantly reduces computational time and resource requirements while preserving the diversity and fidelity of reservoir responses. Ultimately, it supports more efficient uncertainty quantification and scenario evaluation, enhancing the robustness of reservoir development planning.

Methodology

The proposed workflow begins with a conventional uncertainty analysis framework for static reservoir model construction. Key geological parameters—such as facies distribution, petrophysical properties, and structural configurations—are systematically varied to generate an ensemble of static realizations. The combination of facies and property realizations results in a collection of static models, each representing a unique geological scenario. This ensemble forms the basis for uncertainty quantification in volumetrics and serves as input for dynamic simulation workflows. Statistical analysis of the ensemble—such as P10/P50/P90 distributions of volume in place—provides insights into the variability and risk associated with reservoir development.

Each model in the ensemble is then subjected to single-phase streamline simulation, using planned development wells as producers and synthetic injectors to emulate fluid sources. This setup enables the extraction of approximative yet meaningful measures of reservoir connectivity under realistic development scenarios.

Streamline simulation is selected for its computational efficiency and its ability to incorporate fine-scale geological heterogeneity into dynamic reservoir analysis (Batycky, 1997). Unlike full-physics simulations, streamline methods allow rapid evaluation of flow paths and connectivity patterns, making them ideal for screening large ensembles of models.

Traditionally, model screening via streamline simulation focuses on two key metrics: the Shape factor, which quantifies recovery efficiency at 10% of injected fluid, and the Sweep factor, representing the maximum fluid recovery efficiency achieved during the simulation. These metrics are typically visualized and analysed to understand the relationship between fluid transfer efficiency and displacement effectiveness. Figure 1A illustrates the distribution of Shape and Sweep factors across 243 simulated cases, while Figure 1B presents a scatter plot of these efficiency metrics, with each colour denoting a distinct realization.

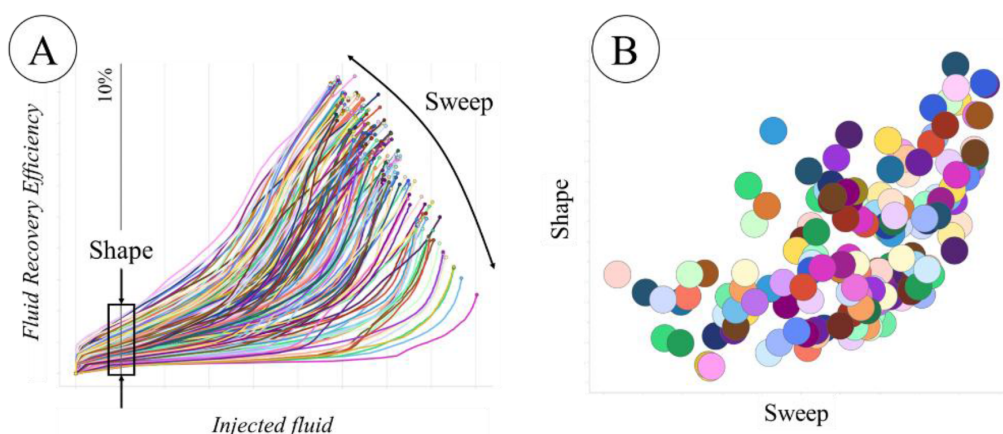


Figure 1—A) Visualization of the Shape and Sweep factor from a collection of 243 streamline simulated cases. B) Scatter plot between efficiency factors (Sweep and Shape), each colour represents a single simulated case

While conventional clustering approaches rely primarily on the correlation between Shape and Sweep factors (Figure-1), the methodology presented here advances this practice by incorporating the full spectrum of outputs from streamline simulation. This includes flow diagnostics, breakthrough times, and volumetric recovery profiles, enabling a more comprehensive and nuanced clustering of geological realizations. Importantly, the methodology here presented is flexible and extensible—additional outputs from streamline simulation can be incorporated to further enrich the clustering process. These may include cumulative fluid production, water production cut, injection rate, injected volumes etc. as reported in Figure-2.

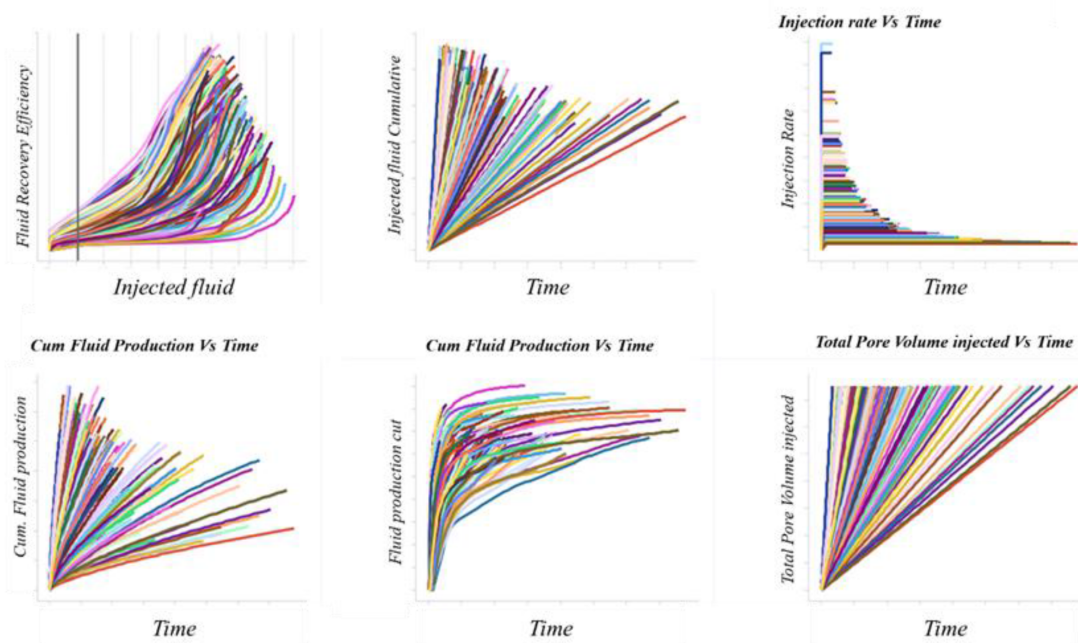


Figure 2—Data visualization Dashboard at field level for all the cases subjected to streamline simulation. A total of 243 different static cases are illustrated with the various line colours.

Furthermore, static volumetric outputs—such as pore volume, net-to-gross ratios, and hydrocarbon-in-place estimates—are reported per realization and integrated with streamline simulation results. These datasets are aggregated and explored using interactive data visualization dashboards, facilitating intuitive model comparison and selection. This integrated approach supports the identification of representative models for subsequent full-physics simulation, significantly reducing computational demands while preserving geological and dynamic diversity (Figure-2).

Static volumetric outputs—primarily volumetric parameters derived from the geological model—and streamline simulation results are systematically reported per realization and subsequently aggregated using interactive data visualization dashboards (Figure 2). This aggregation enables comparative analysis across multiple simulation scenarios and supports decision-making under uncertainty.

To enhance interpretability and facilitate performance evaluation, time series data are visualized concurrently at two hierarchical levels:

- Reservoir Level: Aggregated metrics such as average reservoir pressure, cumulative fluid production, and recovery factor are plotted to provide a comprehensive view of overall reservoir performance trends (Figure-2).
- Well Level: Individual well metrics—including oil and gas production rates, water cut, and bottom-hole pressure—are visualized to assess well-specific dynamics, identify operational anomalies, and support targeted interventions (Figure-3).

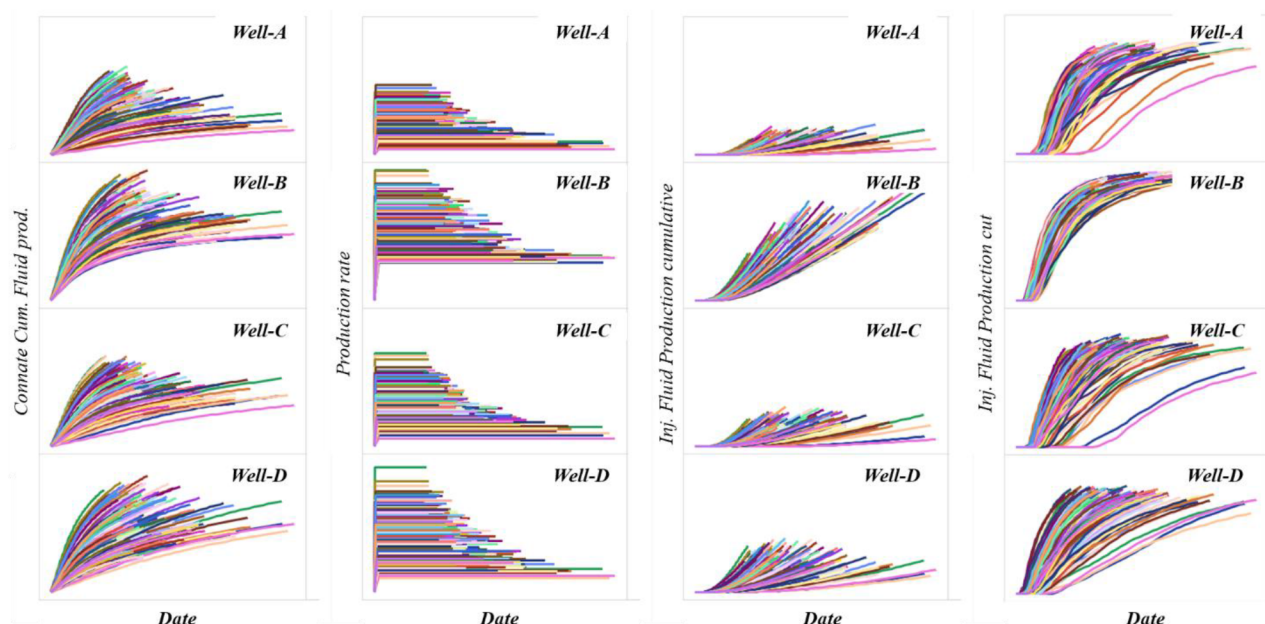


Figure 3—Data visualization Dashboard at well level for four (04) producer wells placed in the reservoir, all wells considered active during the streamline simulation exercise. A total of 243 different static cases are illustrated with the various line colours.

Advanced visualization platforms are employed to generate interactive plots, temporal profiles, and spatial distributions. These tools support dynamic filtering, cross-sectional analysis, and, crucially, the aggregation of results across multiple simulated cases. This integrated visualization approach ensures that KPIs are contextualized within both the static geological framework and the dynamic simulation domain.

Moreover, the coupling of static model volumetrics with dynamic simulation outputs at both field and well levels unlock the potential to explore complex data dependencies—such as spatial production heterogeneity, fluid movement, and inter-well connectivity. This structured integration not only enhances the interpretability of simulation results but also prepares the dataset for future applications in data analytics, including machine learning-based clustering.

Dimensionality Reduction and Clustering Methodology

To facilitate the identification of patterns and groupings within the simulation dataset, a dimensionality reduction step was performed using Principal Component Analysis (PCA). The input dataset comprised 22 variables, including one static volumetric parameter, nine field-level dynamic metrics, and twelve well-level performance indicators. These variables collectively represent a comprehensive characterization of reservoir behaviour across spatial and temporal scales. With this process, successfully reduced the dimensionality to just 3 principal components, Figure-4 illustrate the value of the resulting components and the cumulative contribution of the top five calculates components. Three components capture approximately 92% of the total variability in the data, indicating that they retain most of the original information while significantly simplifying the dataset. This high percentage of explained variance suggests that the underlying structure of the data can be effectively represented in a lower-dimensional space, facilitating easier visualization, interpretation, and further analysis. The reduction also helps mitigate issues related to multicollinearity and overfitting in subsequent modelling tasks.

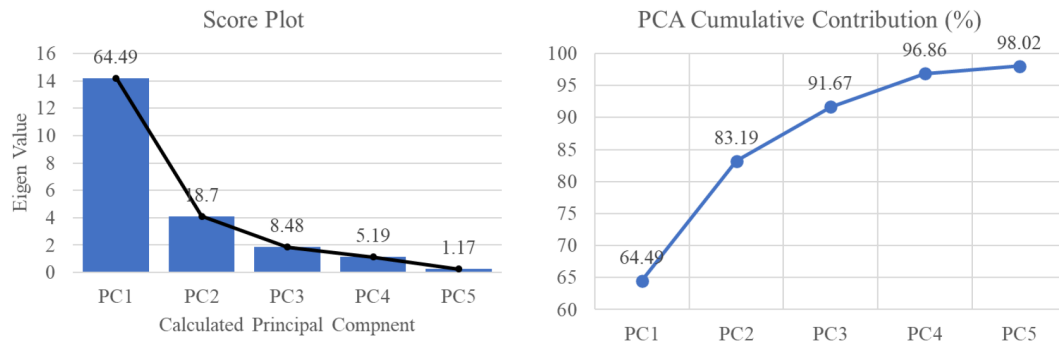


Figure 4—Calculated Eigen values for the resulting Principal Component Analysis (PCA), first three components can be used to approximate about 92% of the data behaviour in a reduced dimensional space.

Prior to PCA, all variables were standardized to ensure uniform scaling and to prevent bias from differing units or magnitudes. PCA was then applied to transform the original high-dimensional dataset into a reduced set of orthogonal components that capture the maximum variance in the data. The number of principal components retained was determined based on cumulative explained variance, ensuring that the reduced representation preserved the essential structure and variability of the original dataset.

The resulting principal components served as input for hierarchical clustering analysis. This approach enabled the classification of simulation realizations into distinct groups based on similarity in reservoir and well-level performance. The combination of PCA and hierarchical clustering provides a robust framework for uncovering latent structures, facilitating scenario comparison, and supporting data-driven decision-making in reservoir management.

Hierarchical Clustering and Representative Case Selection

Following dimensionality reduction via Principal Component Analysis (PCA), hierarchical clustering was applied to the transformed dataset to identify structurally similar groups among the simulation realizations. The clustering process was performed using agglomerative linkage, which iteratively merges data points based on their pairwise distances in the reduced feature space. Resulting dendrogram is showcased in [Figure-5](#) whereas all similar cases are grouped based on their similarity or distance within the PCs from one another, using hierarchical clustering. The process starts by treating each item as its own cluster. Then, the algorithm successively merges the closest pairs of clusters based on Euclidean distance and complete.

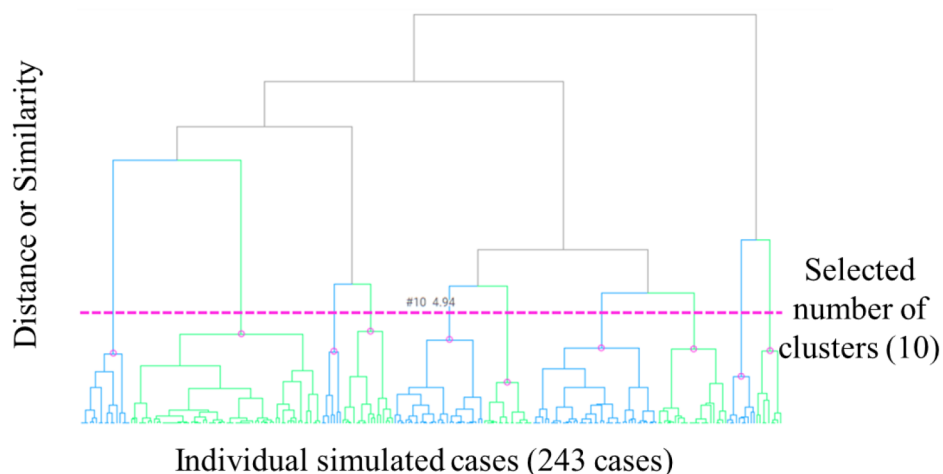


Figure 5—Dendrogram used for Hierarchical clustering, case grouping along the horizontal axis is based on their similarity or distance within the calculated principal components.

To determine the optimal number of clusters, the elbow method was employed. This technique involves plotting the within-cluster sum of squares (WCSS) against the number of clusters and identifying the point at which the rate of decrease sharply changes—commonly referred to as the "elbow." This inflection point indicates a balance between model complexity and explanatory power, guiding the selection of a clustering configuration that captures meaningful variation without overfitting.

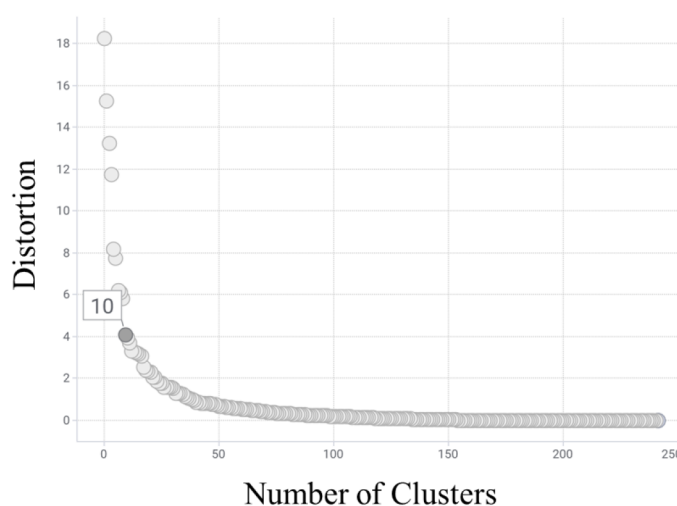


Figure 6—Elbow plot illustrating the "Elbow" point at which the rate of decrease in distortion sharply changes.

Once the optimal number of clusters was established, the dataset was categorized accordingly. Each simulation realization was assigned to a cluster, enabling structured comparison across groups. This categorization was subsequently used to select representative cases from each cluster. Representative cases were chosen based on their proximity to the cluster centroid in PCA space, ensuring that selected realizations reflect the central tendencies and dominant behaviours of their respective groups.

This clustering-based selection process supports scenario reduction, enhances interpretability, and provides a robust foundation for further analysis, including sensitivity studies, uncertainty quantification, and data-driven forecasting. The strategic value of this selection lies in its ability to establish lean simulation workflows. By focusing computational resources on a reduced set of representative cases, engineers can perform targeted analyses that preserve the diversity of reservoir responses while significantly reducing simulation time and cost. This approach is particularly beneficial for sensitivity studies, scenario screening, and uncertainty quantification, where exhaustive simulation of all realizations may be impractical.

Results

The primary output of the clustering methodology is the assignment of a cluster classification label to each static geological realization within the ensemble (Figure-7). These labels reflect groupings based on dynamic performance derived from streamline simulation outputs, such as flow connectivity, sweep efficiency, and drainage patterns. The clustering process enables the identification of key representative realizations within each group—models that not only capture the full range of volumetric properties but also reflect the diversity of dynamic reservoir responses observed across the ensemble.

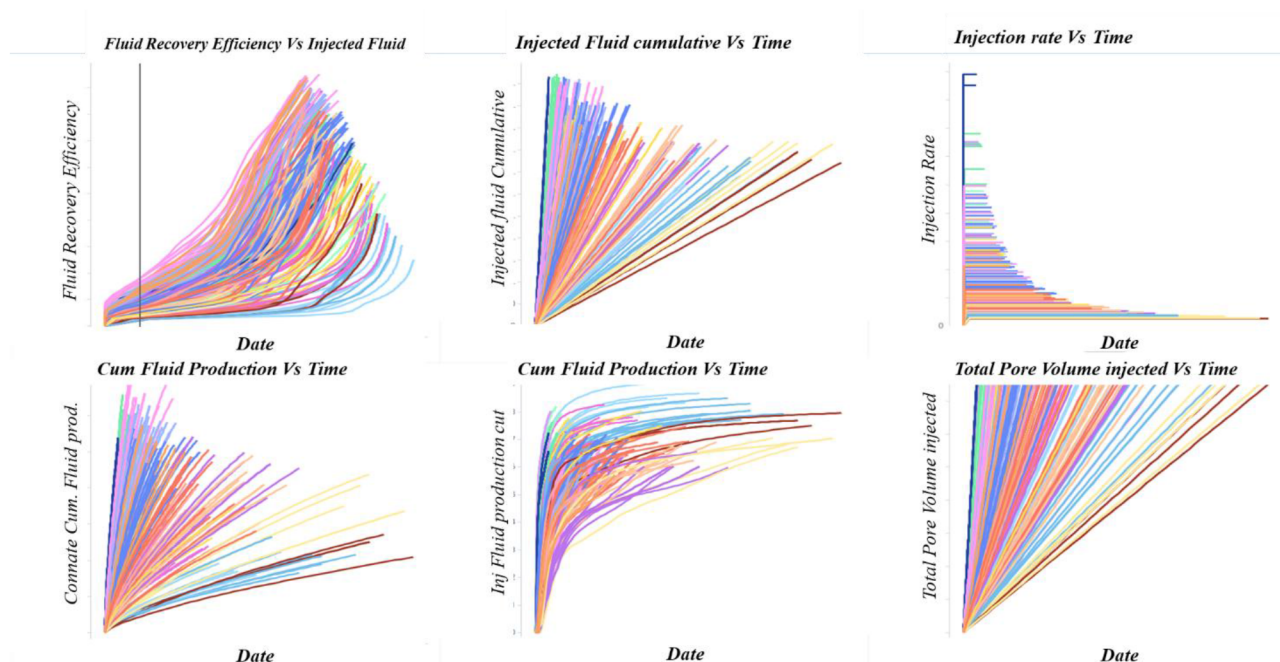


Figure 7—Visualization dashboard displaying the 243 simulated cases after clustering. The time series are color-coded according to their respective hierarchical cluster assignments.

Representative realizations are selected based on their proximity to the cluster centroid in the reduced PCA space, ensuring that each chosen model embodies the central tendencies of its group. This selection is critical for reducing the number of cases required for subsequent full-physics simulations, allowing for a lean yet comprehensive simulation strategy that maintains fidelity while improving computational efficiency.

To support interpretation and decision-making, an interactive visualization dashboard is employed. Each realization is color-coded according to its assigned cluster label, enabling intuitive exploration of the ensemble structure. The dashboard facilitates dynamic filtering, cross-sectional analysis, and the generation of a shortlist of representative cases, which are then prioritized for detailed simulation and performance evaluation. This integrated workflow—combining streamline simulation, clustering, and visualization—provides a robust framework for scenario reduction, uncertainty quantification, and strategic planning in reservoir development.

A key outcome of the proposed methodology is the substantial reduction in the number of simulation cases required for full-physics modelling. Initially, the study considers an ensemble of 243 static geological realizations (as illustrated in Figure 1), each representing a plausible scenario of subsurface heterogeneity and volumetric uncertainty. Running dynamic simulations for all these cases would be computationally prohibitive and time intensive.

Through the application of streamline simulation and hierarchical clustering, the ensemble is systematically analysed to identify patterns in flow behaviour and connectivity. This enables the classification of realizations into distinct groups based on dynamic similarity. From each cluster, a small number of representative cases are selected models that effectively capture the volumetric range and dynamic variability of their respective groups.

As a result, the simulation scope is narrowed down to just 10 representative cases. This targeted selection allows for a lean simulation strategy that maintains analytical rigor while significantly reducing computational demands. By focusing on the most informative and diverse realizations, the approach enables faster performance assessments, supports uncertainty quantification, and facilitates the generation of additional insights into reservoir behaviour.

This reduction in the number of simulation cases—from an initial set of 243 to a focused subset of 10 representative realizations—not only accelerates the simulation phase but also significantly enhances the overall efficiency of the reservoir modelling workflow. By minimizing the computational burden associated with full-physics simulations, the methodology enables faster turnaround times for performance evaluation, scenario comparison, and sensitivity analysis.

The lean simulation strategy makes the workflow more practical and scalable, particularly in iterative modelling environments where multiple rounds of updates, calibrations, and optimizations are required. It allows reservoir engineers to rapidly test hypotheses, refine development strategies, and respond to new data or operational constraints without being hindered by excessive computational overhead.

Moreover, the ability to concentrate efforts on the most informative and diverse cases improves the quality of insights derived from the modelling process. It supports more robust scenario screening, facilitates uncertainty quantification, and strengthens the foundation for data-driven decision-making. In field development planning, this translates into more confident forecasting, optimized resource allocation, and reduced risk in investment decisions. Ultimately, the approach balances analytical rigor with operational practicality, enabling high-resolution reservoir understanding while maintaining agility in project execution.

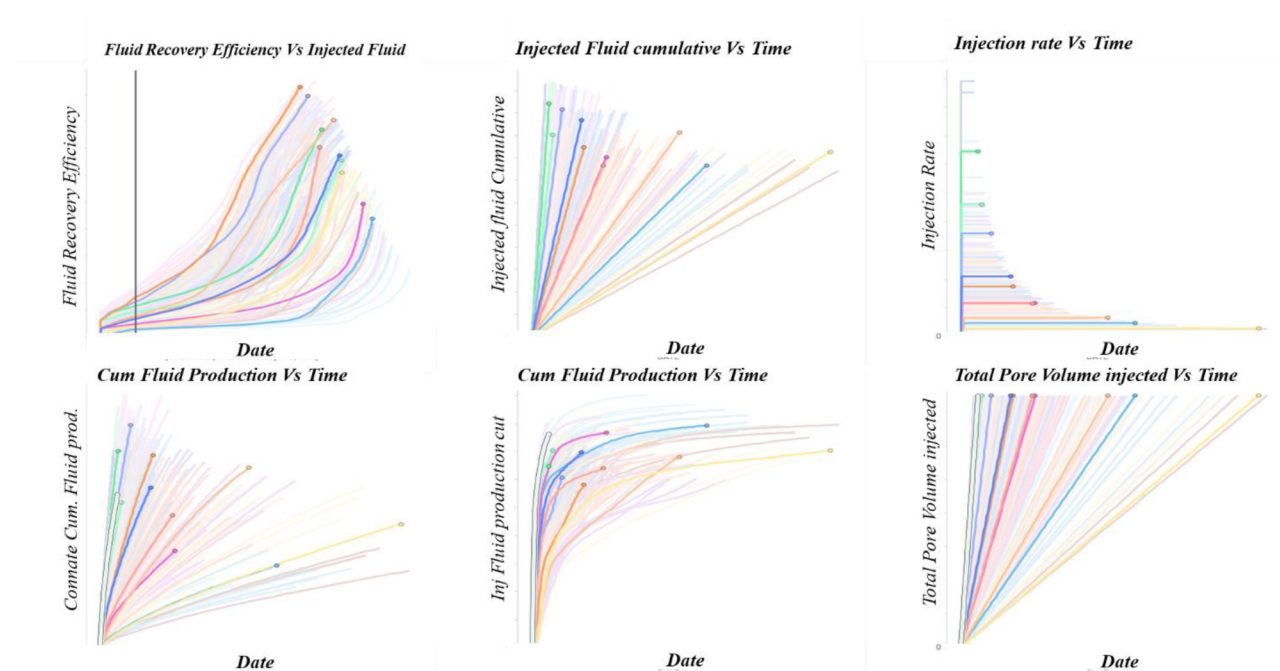


Figure 8—Data visualization Dashboard for the 243 simulated cases after clustering. A total of 10 representative cases are highlighted with the various line colours. Highlighted cases are selected for full physics dynamic simulation.

Discussion

The clustering methodology applied in this study has yielded a structured and interpretable classification of the 243 static geological realizations, each labelled according to its dynamic performance characteristics derived from streamline simulation outputs. These labels reflect groupings based on critical flow-related metrics such as sweep efficiency, drainage patterns, and inter-well connectivity.

This classification has proven efficiency in identifying key representative realizations within each cluster. These representatives are selected based on their proximity to the cluster centroid in the reduced PCA space, ensuring that they embody the central tendencies of their respective groups. Importantly, these models capture both the volumetric diversity and the dynamic variability present in the full ensemble, making them ideal candidates for further full-physics simulation.

The use of an interactive visualization dashboard has further enhanced the interpretability and usability of the clustering results. By color-coding each realization according to its cluster label, the dashboard enables intuitive exploration of the ensemble structure. Users can dynamically filter and analyse realizations, perform cross-sectional comparisons, and generate a shortlist of representative cases for detailed simulation. This visual and interactive component supports real-time decision-making and facilitates communication across multidisciplinary teams.

One of the most impactful outcomes of this methodology is the reduction in simulation scope—from 243 initial cases to just 10 representative realizations. This reduction translates into a substantial gain in computational efficiency, allowing for faster turnaround times in performance evaluation, scenario screening, and sensitivity analysis. It also makes the workflow more scalable and practical, particularly in iterative modelling environments where multiple rounds of updates and optimizations are required.

By concentrating simulation efforts on the most informative and diverse cases, the approach enhances the quality of insights derived from the modelling process. It supports robust uncertainty quantification and strengthens the foundation for data-driven decision-making in field development planning. The ability to maintain analytical rigor while reducing computational cost is especially valuable in contexts where resources are limited, or rapid assessments are needed.

Ultimately, this integrated workflow—combining streamline simulation, PCA-based dimensionality reduction, hierarchical clustering, and interactive visualization—offers a balanced and efficient strategy for reservoir modelling. It enables high-resolution understanding of subsurface behaviour while preserving agility in execution, making it a powerful tool for modern reservoir management.

Final quality check is performed by plotting static and dynamic indicators (from the streamlines) and colour coding the graph with the defined clusters. Figure-9 illustrated this relationship and clearly confirms that case selection is regularly spaced and strategically covering a good portion of the variability of the static realizations.

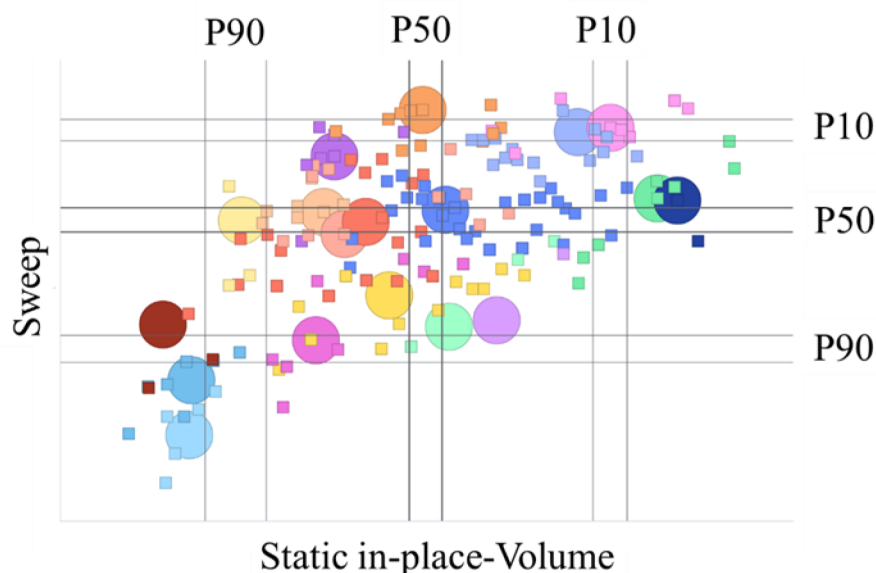


Figure 9—Scatter plot for X-Axis (Volume-in-Place initially present in the reservoir and Y-Axis (Sweep Factor) Representing the fraction of the reservoir volume that has been swept by the injected fluid. Colour code is as per cluster defined and big dots represents the shortlisted cases.

Conclusions

The geological screening process has been significantly enhanced through the integration of machine learning techniques and principal component analysis (PCA), enabling a more robust and scalable

approach to uncertainty quantification. By applying PCA to a high-dimensional dataset composed of static and dynamic reservoir descriptors, the methodology effectively retains key variances—particularly those associated with connectivity and flow behaviour—while reducing dimensionality for more efficient analysis.

Machine learning algorithms, particularly hierarchical clustering, are then applied to the reduced dataset to identify patterns and groupings among geological realizations. This data-driven classification facilitates the selection of representative models that capture the full spectrum of geological and dynamic variability, supporting scenario reduction and targeted simulation strategies.

To ensure the reliability and transparency of the process, quality control is performed using interactive visualization dashboards. These dashboards allow users to explore the ensemble in real time, with cluster labels color-coded for intuitive interpretation. Users can dynamically filter, compare, and validate realizations based on volumetric properties, connectivity metrics, and flow responses.

This integrated approach not only improves decision-making by providing clearer insights into reservoir behaviour but also enhances workflow efficiency and modelling accuracy. By narrowing the focus to the most informative cases, it significantly reduces computational time and cost, making it practical for iterative analysis, rapid screening, and strategic planning in reservoir development.

The presented advanced static model screening workflow, utilizes hierarchical clustering and principal component analysis (PCA), has demonstrated significant improvements in the efficient use of simulation resources. Key conclusions from this study include:

- Static and dynamic integration using streamline simulation.
- Creation of interactive data visualization dashboards to expedite analysis and link static and dynamic parameters in a common platform
- Enhanced Model Selection: This approach reduced redundancy and focused computational efforts on the most impactful scenarios.
- Dimensionality Reduction: By applying PCA, the complexity of the dataset was significantly reduced while retaining the essential variability and features of the geological models.
- Resource Optimization: The combined use of hierarchical clustering and PCA led to a more streamlined workflow, optimizing the allocation of computational resources.
- Improved Decision-Making: The refined geo-screening process provided clearer insights into reservoir behaviour, uncertainty managements enhancing the decision-making process for reservoir management and development strategies. The ability to focus on key models and parameters improved the reliability of predictions and planning.

Recommendations and way forward

The methodology presented in this work was specifically developed and implemented during the greenfield stage of reservoir development. At this early phase, when limited production data is available, the primary objective was to systematically identify and select representative reservoir models that encompass the full range of possible scenarios in terms of both reservoir connectivity and volumetric characteristics. This was achieved through the use of streamline-based simulation techniques, which offer a powerful and efficient means of visualizing and quantifying fluid flow paths and reservoir heterogeneity.

Beyond its initial application at the greenfield stage, the methodology also holds significant potential for use in later phases of reservoir management. Specifically, it can be employed to refine the selection of analogous reservoir cases during history matching, where observed production data is used to calibrate simulation models. By identifying models with similar connectivity and volumetric profiles, the methodology enhances the efficiency and accuracy of the history matching process, ultimately contributing to improved forecasting and reservoir performance evaluation.

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